

The origin, dispersal and distribution of the American mink *Mustela vison* in Poland

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Brzeziński M. and Marzec M. 2003. The origin, dispersal and distribution of the American mink *Mustela vison* in Poland. *Acta Theriologica* 48: 505–514.

In Poland, feral population of the American mink *Mustela vison* Schreber, 1777, originating from the immigrants from Belarus and Lithuania, as well as from Polish fur farm escapees, started to develop in the beginning of 1980s in the north-east and north-west of the country. According to questionnaires sent to hunters, up to the end of 1980s mink colonized 9 out of 16 Polish provinces. Within next decade mink were observed in next 6 provinces. In Warmińsko-Mazurskie and Podlaskie provinces (north-eastern Poland) mink presence has been confirmed by over 90% of hunting units, whereas in the north-west of the country by over 50% of respondents. In southern provinces mink are still uncommon but they successfully enlarge geographical range. During the 20-year period, mink have colonized over half of the Polish territory. Colonization of particular regions was correlated in time with a decline of muskrat and waterfowl populations. From the beginning of 1980s to the half of 1990s, in north-eastern provinces hunting bag of muskrat decreased 10-fold, whereas in other parts of Poland the number of hunted muskrats was more or less stable. Despite the lack of complex ecological studies on the mink impact on their prey, it is highly probable that this introduced carnivore negatively affects populations of semi-aquatic birds and mammals.

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Key-words: *Mustela vison*, geographic range, expansion, Poland

Introduction

The American mink *Mustela vison* Schreber, 1777, a semi-aquatic mustelid from North America, is a valuable fur-bearing animal, which has been bred on fur farms since the end of the 19th century (Dunstone 1993). In the 1920s and 1930s mink farms were established in many European countries. Escapees from fur farms bred in the wild initiating feral populations which spread over large areas in Britain, Fennoscandia, central and eastern Europe (in Russia and Belarus minks were also released to the wild as game animals). In the second half of the 20th century areas invaded and populated by mink expanded on the whole continent. Such countries as Sweden, Scotland, Norway and Finland were colonized during 20–30 year periods (Gerell 1967, Cuthbert 1973, Bevanger and Henriksen 1995, Kauhala 1996). In Poland mink farms developed in the 1950s (Lisiecki and Sławoń 1980) but mink colonization begun 30 years later, in the early 1980s. The expansion of this species

in Poland has not been well studied. Some of the first mink records were presented by Ruprecht *et al.* (1983), Romanowski *et al.* (1984), Żurowski and Kammler (1987), Ruprecht (1996) but they did not show the overall picture of mink dispersal in the country.

The impact of expanding mink populations on native fauna has been studied in several European countries (eg: Gerell 1985, Woodroffe *et al.* 1990, Andersson 1992, Craik 1997, Aars *et al.* 2001). These studies revealed that, locally, populations of some semi-aquatic mammals and water birds are vulnerable to mink predation, especially soon after the mink colonization of a certain area, however, the long-term ecological consequences of the mink presence in aquatic habitats are usually unknown. It must be taken into account that also in Poland some of potential mink prey, such as ducks, coots *Fulica atra*, water voles *Arvicola terrestris* and muskrats *Ondatra zibethicus*, may decline in numbers due to the predation of American mink.

This paper describes the mink history in Poland and its present distribution. Available data on the mink expansion and the hunting bag of waterfowl and muskrat enable to discuss the mink impact on some mammals and birds inhabiting water ecosystems. However, due to the lack of long-term field studies, presented data should be treated with proper caution.

Material and methods

The history of American mink dispersion in Poland is analysed on the basis of available literature and questionnaires which were sent to units of the Polish Hunting Association in 1998. Hunters were asked to answer several questions concerning American mink presence in particular hunting districts: does mink inhabit local water bodies, when it has been recorded for the first time, is it common in the area (at least 10 records in one year), are there any visible changes in muskrat and waterfowl numbers? Questionnaires were sent back from 727 hunting units. Results are presented for 16 administrative provinces. Game bags from the period 1981–1997 were used to compare the changes of muskrat and waterfowl numbers in different regions of Poland.

Results

The origin of Polish mink population

In Poland, mink farming begun in 1953 (Lisiecki and Sławoń 1980) and feral mink recorded in 1950s and 1960s were undoubtedly fugitives from farms. In that period of time, despite single reports of feral mink females rearing cubs in the wild (Ruprecht 1996), there was no evidence of any wild living mink population enlarging its range in Poland. In the 1960s feral mink population was already abundant in Belarus, not far from the Polish eastern border, and in the 1980s mink population in that country was estimated for about 50 000 individuals (Sidorovich 1991). The nearby Lithuania was colonized by American mink in the 1970s and in the first half of 1980s (Mickevicius and Baranauskas 1992). Thus, one may suspect that the mink population, at least in north-eastern Poland, originated from immigrants coming from the east. The localization of mink records in the first stage

of expansion in Poland indicates that mink extended its geographical range to the west and south-west. It must be noted, however, that in the beginning of the 1980s mink were also recorded in north-western Poland. Due to the long distance to the Belorussian and Lithuanian populations, these animals could not be immigrants arriving from the east. Most probably they escaped from mink farms and successfully started to breed in the wild (in north-western Poland mink farms were more numerous than in other parts of the country; in 1997, 10 out of 19 breeding farms were located in north-western Poland – Dąbrowska and Krawczyk 1997). Thus, besides the main expanding mink population, some local populations also came into existence, established by animals which escaped from fur farms.

The distributional history of the American mink in Poland

In Poland, first observations of American mink in the wild come from the end of the 1950s. During the two following decades new records of this carnivore were rather scarce (Ruprecht 1996). Until the late 1970s American mink was recorded in scattered sites in different parts of the country. Most of these sites were located in northern Poland. Sporadic observations suggest that in this period of time wild living mink populations were not abundant and observed animals were probably single fugitives from mink farms. The only exception could be mink observed in 1972–1973 in Białowieża Primeval Forest (Ruprecht *et al.* 1983) and in 1976 and 1978 in Knyszyńska Primeval Forest (Romanowski *et al.* 1984), perhaps first immigrants from the Belarussian population.

In the beginning of the 1980s, observations of wild living mink became more numerous. Ruprecht *et al.* (1983) presented records of American mink in Białowieża Primeval Forest in years 1980–1982. Many new observations were made in 1980–1987 in other parts of north-eastern Poland (Ruprecht 1996). In that time first mink were sighted at the upper Narew River, Supraśl River, Biebrza River and at Lake Wigry (Romanowski *et al.* 1984). First observations in Mazury Lakeland come from Lake Beldany, where minks were recorded in 1982 (Ruprecht 1996). In 1984–1987 many new sites inhabited by mink in Piska Primeval Forest (Mazury Lakeland) were described by Żurowski and Kammler (1987). In 1983 and 1984 Schmidt and Kaszuba (1985) observed American mink at Lake Družno near Elbląg, close to the Baltic coast. In the 1980s in Warmińsko-Mazurskie province and Podlaskie province many wild living mink were observed by hunters (Wolny 1987, Żebrowski 1987). In the same period of time single American mink were recorded in Mazowieckie province, not far from Warsaw (Romanowski *et al.* 1984). In the middle of the 1980s some observations of these mustelids were made in Zachodnio-Pomorskie province (eg Wałeckie Lakeland), however they were not as numerous as in north-eastern Poland (Włodek 1993, Ruprecht 1996).

Increasing number of mink records suggest that in the 1980s American mink begun its expansion in Poland and first stable and breeding populations were established in north-east and probably also in north-west of the country. In the following years the geographical range of mink expanded mainly to the west and

south-west. Questionnaires received from hunters support this scenario of dispersion. In 1970–1979 American mink was observed sporadically only in five northern and north-eastern provinces (Zachodnio-Pomorskie, Pomorskie, Warmińsko-Mazurskie, Podlaskie, Mazowieckie). In the 1980s mink became more numerous in these regions and they appeared in the next four, central and western provinces (Kujawsko-Pomorskie, Wielkopolskie, Lubuskie, Dolnośląskie) (Fig. 1a). Up till 1998 mink was recorded in 15 provinces, however in the south of the country the number of observations was still very small. In southern provinces first records come from the middle of the 1990s.

The geographical range up to 1998 and further mink expansion in Poland

According to the data obtained from questionnaires, mink are most common in north-eastern Poland. In Warmińsko-Mazurskie province and in Podlaskie province their presence has been confirmed by over 90% of hunting units (Fig. 1b). Over 50% of positive answers were sent from Pomorskie province and Lubuskie province. In southern provinces mink are still uncommon and there are only single reports on mink occurrence in this area.

Majority of hunters (58–82%) from Warmińsko-Mazurskie, Podlaskie, Pomorskie and Lubuskie provinces state that mink is common in their hunting units. In all other provinces, most of respondents suggest that mink are rather uncommon. In central and southern Poland, where mink have been recorded only in a few sites, observations of this carnivore are still very rare.

An analysis of consecutive mink records between 1980–1998 shows the main directions of its dispersion in Poland. According to questionnaires, in the 1980s American mink settled in lakelands and in the Narew River basin, in the north-eastern Poland (Warmińsko-Mazurskie and Podlaskie provinces), and these results correspond to earlier publications (Ruprecht *et al.* 1983, Romanowski *et al.* 1984, Ruprecht 1996). In the following years mink enlarged their range mainly to the west, reaching the lower course of Wisła River (Pomorskie and Kujawsko-Pomorskie provinces), and to the south, reaching Bug River and the middle course of Wisła River (Mazowieckie province). Wild living mink appeared also in north-western Poland, far away from the eastern population. Up to 1998, mink expanded to the south upstream main rivers, such as Odra, Warta, Pilica and Wisła, and they have already reached rivers in Polish uplands. In last years mink were observed at catchments of the upper Wisła River (Świętokrzyskie and Małopolskie provinces) and in lower course of San River (Podkarpackie province). During the 20-year period, mink have colonized over half of the Polish territory and is still expanding to the south of the country.

In Poland, mink densities in particular habitats are unknown, except of two areas where mink populations have been studied. In Białowieża Primeval Forest mink densities were highest at medium-sized rivers – up to 7.5 individuals/10 km of river course (Sidorovich *et al.* 1996). Mean densities varied from 1.9 individuals/10 km at small streams to 6.3 individuals/10 km at medium-sized rivers. Mink

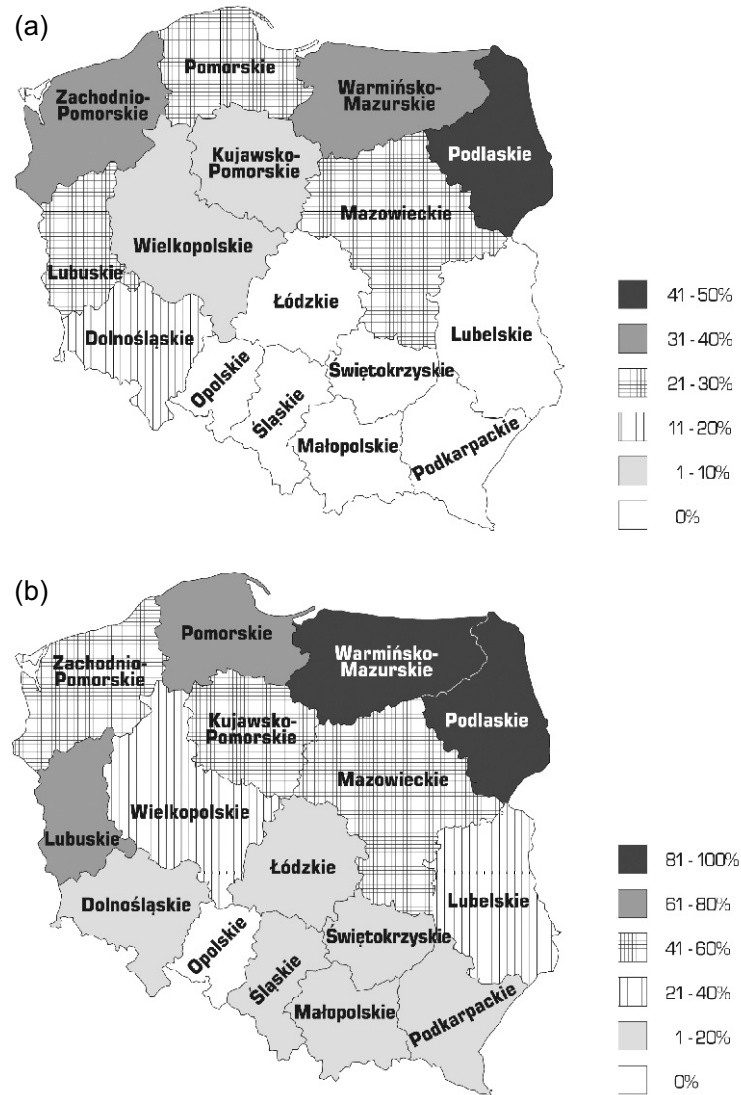


Fig. 1. (a) First records of American mink in provinces. Percentage of questionnaires reporting mink presence in which first mink were recorded before 1990. (b) American mink abundance in provinces. Percentage of questionnaires in which hunters reported mink presence in a hunting district.

densities recorded in Mazurian Lakeland were higher than in Białowieża Primeval Forest, at five lakes studied, their multi-annual average densities varied from 7 to 10 individuals/10 km of lake shore (M. Brzeziński, in prep.) However, at some lakes early-spring densities reached 14 individuals/10 km. In northern Poland numerous lakes provide excellent habitats for mink and probably these suitable habitats facilitated very fast colonization of northern provinces. The large number of mink

observations made in lakeland areas by hunters may also suggest that in this part of the country mink densities are still higher than in other regions.

Mink impact on prey populations

Data on ducks, coots and muskrat obtained from questionnaires (subjective view of hunters) and from Polish Hunting Association (hunting bag) have to be treated with caution and can not provide final conclusions about mink impact on populations of these animals. However, while discussing the role of American mink in colonized areas, these data should be analysed and taken into consideration. Most

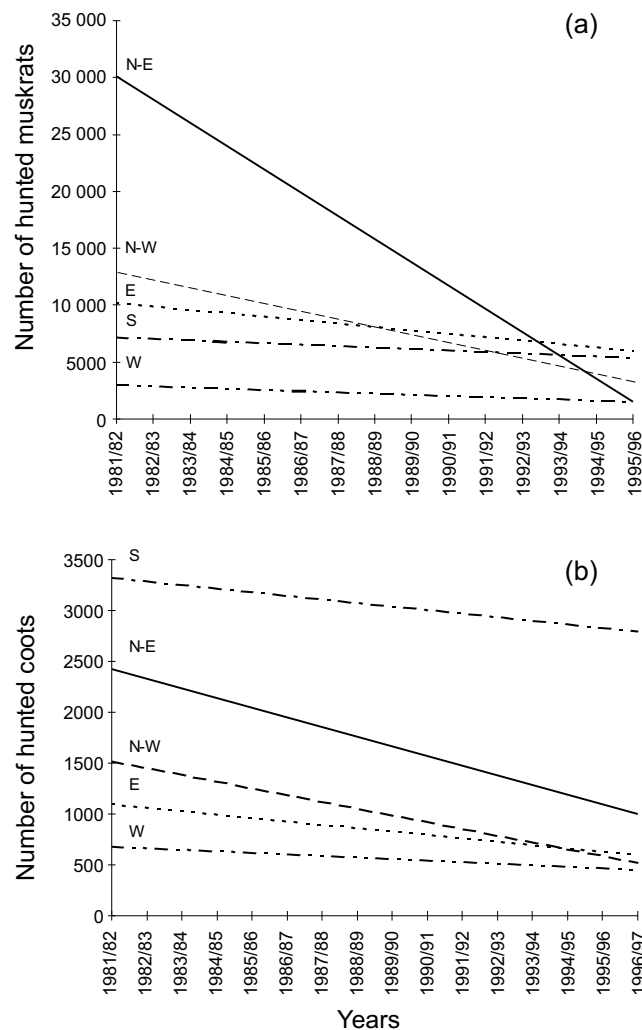


Fig. 2. (a) Hunting bag of muskrats in five regions of Poland in years 1981–1996 (trend lines). (b) Hunting bag of coots in five regions of Poland in years 1981–1997 (trend lines).

of hunters who filled up our questionnaires and recorded mink presence in their hunting units, informed of a decline in waterfowl and muskrat populations. In areas where first mink were observed in 1970s, 1980s and in the beginning of 1990s over 80% of hunting units reported negative changes in numbers of ducks, coots and muskrats. According to the information obtained from hunters, populations of these game animals were stable in most of areas uninhabited by mink. However, it must be noted, that some hunters observed changes in muskrat and waterfowl numbers in habitats where mink did not occur. Muskrat population decline was reported from 26.2% of such hunting units, and waterfowl (mainly mallard *Anas platyrhynchos*) population decline in 12.7%. It means that also other factors than mink predation have to be considered.

During the last 20 years, as revealed by statistics of the Polish Hunting Association, muskrat, mallard and coot game bag declined in the whole country, probably reflecting the decline in population numbers of these animals. In different regions of Poland changes were not the same (Fig. 2). The most significant decline of the game bag took place in north-eastern and in north-western Poland, whereas in other regions changes were not very obvious. A detailed analysis of the numbers of muskrats hunted in Warmińsko-Mazurskie province and northern part of Podlaskie province (NE Poland) shows that the muskrat population rapidly begun to decline in the second half of the 1980s (Fig. 3). This process was strongly correlated in time with mink colonization of the area. Reports sent by hunters and game bag statistics strongly support supposition that American mink is responsible for the decline of muskrat, mallard and coot populations in the areas invaded by this carnivore, at least in the first years of colonization. However, long-term ecological relations between mink and their prey need additional data, other than these obtained from questionnaires and Polish Hunting Association.

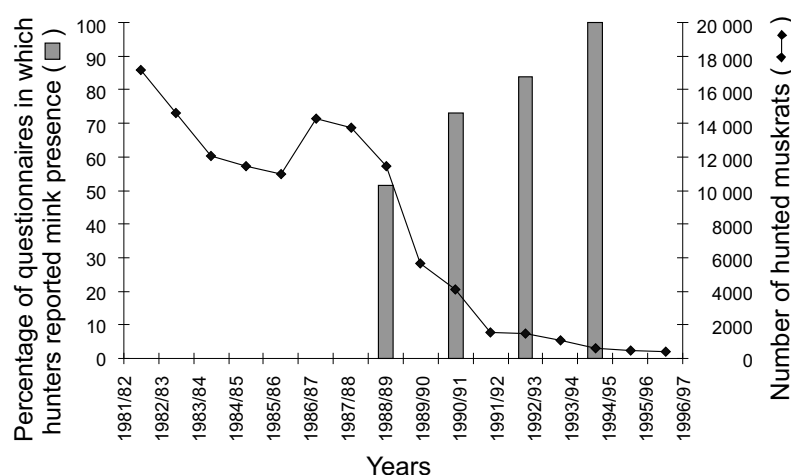


Fig. 3. Hunting bag of muskrats in NE Poland (Warmińsko-Mazurskie province and northern part of Podlaskie province) and the number of mink records in the hunting units of the same region (100% = all hunting units in which mink was recorded up to 1998).

Discussion

Despite the fact that first reports on feral mink in Poland come from the 1950s and the 1960s, it can be stated that the true colonization of the country begun in the beginning of the 1980s. Polish population originated from eastern populations established in Belarus, as well as from escapees from fur farms, most of which were located in north-western Poland. Within 10 years, up to the beginning of the 1990s, mink colonized lakelands in northern Poland and begun its further dispersion. During the next decade it moved far to the south reaching San River basin (Podkarpackie province), upper Wisła River basin and upper Odra River basins (Świętokrzyskie, Małopolskie, Śląskie, Dolnośląskie provinces). One may suspect that within the next few years mink will reach mountainous regions in southern Poland and colonize the whole country. Although mink expansion in Poland started much later than in many other European countries, it was as rapid as in Scotland, Fennoscandia or Baltic states.

In regions colonized in the 1980s, mink populations are rather stable, reaching high densities, especially in good quality habitats, such as eutrophic lakes. In recently colonized areas mink are probably still rare, however, due to the high productivity of these animals, mink densities in newly colonized habitats may increase rapidly (Sidorovich 1993). The success of mink expansion in Poland, similarly to other parts of Europe, is due to several factors, of which the most important is the mink's ability to occupy an unexploited ecological niche. The high environmental plasticity of mink and their generalistic food habits enable inhabiting various aquatic habitats. Mink can effectively breed in the wild and the human impact, even with hunting and intensive trapping, is never strong enough to exterminate expanding populations. Examples from other countries show that mink dispersion, once triggered, can last for many years, and animals can increase densities until they reach the environmental capacity (Gerell 1967, Bevanger and Henriksen 1995, Kauhala 1996). Thus, there can be no doubt that feral mink became a permanent member of Polish fauna.

The main ecological consequence of mink dispersion in Europe is their impact on prey populations. Birds and mammals inhabiting aquatic ecosystems are most vulnerable to the mink predation. In the short period of time, especially soon after the colonization of a new area, mink can considerably affect populations of muskrat, coot, mallard and other waterfowl. Gerell (1985), Niemimaa and Pokki (1990), Woodroffe *et al.* (1990), Andersson (1992), Halliwell and Macdonald (1996) showed a significant role of American mink in reducing numbers of waterfowl and water vole populations. Although some authors neglect the role of American mink in decreasing muskrat densities, arguing that changes of their numbers may result from many different ecological factors (Ruprecht 1996), the tenfold decline in muskrat hunting bag in north-eastern Poland in years 1980–1997, strongly supports the theory that mink is a predator which can affect populations of this rodent. Decline in muskrat numbers and mink dispersion in north-eastern Poland were correlated in time. Balerstet *et al.* (1990) who studied muskrat population at

Drużno Lake, reported rapid decline of muskrat numbers soon after mink had colonized the area in the end of the 1980s. Other factors, such as droughts, drainage, epizootic outbreaks and intraspecific competition also affect muskrat dynamics (Le Boulenge-Nguyen 1981, Clark and Kroeker 1993) but mink predation can significantly deepen the decrease in numbers of these semi-aquatic rodents. In Canada both muskrat and mink populations cycle with a period of 8–9 years, and usually mink tend to lag 1–2 years behind muskrat (Viljugrein *et al.* 2001). In this natural predator-prey relationship mink predation does not trigger muskrat cycles and does not lead to permanent decline of rodent populations. However, in regions of Europe recently colonized by mink, the scenario of predator-prey relationship between mink and muskrat may be slightly different. The role of mink predation in shaping muskrat dynamics can be significant, similar as it has been recorded for water vole (Woodroffe *et al.* 1990, Halliwell and Macdonald 1996, Aars *et al.* 2001).

Mink impact on waterfowl can also be significant, especially during the breeding season. In some areas the number of successful broods of ducks, coots, gulls, sterns and waders can be seriously limited by this predator (Oring *et al.* 1983, Burness and Morris 1993). In Fennoscandia, Gerell (1985), Niemimaa and Pokki (1990), and Andersson (1992) observed that many bird colonies vanished due to mink abundance. Locally, mink predation can exterminate breeding birds, but some species can avoid this predator by moving to new, inaccessible breeding sites (Dunstone 1993, Kilpi 1995). Thus, it would be difficult to deny that a decline of mallard and coot populations in many regions of Poland, could not have been affected by the expanding mink. However, the extend of mink predation and its importance to the long term dynamics of waterfowl, need further, more detailed studies.

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Received 7 February 2002, accepted 5 May 2003.

Associate Editor was Andrzej Zalewski.